

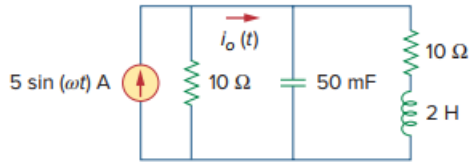
Problem

In the circuit of Fig. 9.47, find i_o when:

(a) $\omega = 1 \text{ rad/s}$

(b) $\omega = 5 \text{ rad/s}$

(c) $\omega = 10 \text{ rad/s}$



Step-by-step solution

Step 1 of 15

Refer to Figure 9.47 in the text book.

Consider the value of the current source $i(t)$.

$$i(t) = 5 \sin \omega t \text{ A}$$

Represent current source $i(t)$ in phasor form.

$$I = 5 \angle 0^\circ \text{ A}$$

Therefore, the value of the source in phasor form is $5 \angle 0^\circ \text{ A}$.

Step 2 of 15

(a)

For $\omega = 1 \text{ rad/s}$

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute 1 rad/s for ω and 2 H for L .

$$\begin{aligned} X_L &= j(1)(2) \\ &= j2\Omega \end{aligned}$$

Therefore, the value of the reactance X_L is $j2\Omega$.

Step 3 of 15

Calculate the value of the reactance due to capacitance.

$$X_c = \frac{1}{j\omega C}$$

Substitute 1 rad/s for ω and 50 mF for C .

$$\begin{aligned} X_c &= \frac{1}{j(1)(50 \times 10^{-3})} \\ &= -j20 \Omega \end{aligned}$$

Therefore, the value of the reactance X_c is $-j20 \Omega$.

Step 4 of 15

Draw the circuit diagram in phasor representation as shown in Figure1.

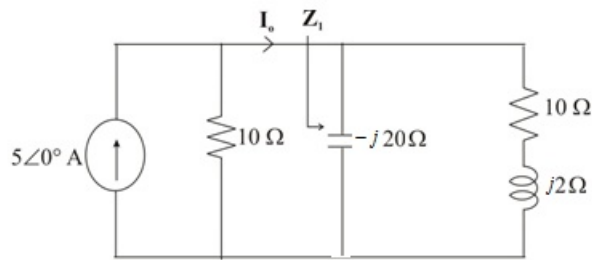


Figure 1

Step 5 of 15

Calculate the value of the impedance Z_1 .

$$\begin{aligned} Z_1 &= -j20 \parallel (10 + j2) \\ &= \frac{(-j20)(10 + j2)}{(-j20) + (10 + j2)} \\ &= \frac{40 - j200}{10 - j18} \\ &= 9.433 - j3.0288 \Omega \end{aligned}$$

Step 6 of 15

Calculate the value of the current $\mathbf{I}_o$ using current division rule.

$$\mathbf{I}_o = (5\angle 0^\circ) \left(\frac{10}{\mathbf{Z}_1 + 10} \right)$$

Substitute $9.433 - 3.0288j\Omega$ for $\mathbf{Z}_1$.

$$\begin{aligned}\mathbf{I}_o &= (5\angle 0^\circ) \left(\frac{10}{9.433 - j3.0288 + 10} \right) \\ &= (5\angle 0^\circ) \left(\frac{10}{19.433 - j3.0288} \right) \\ &= (5\angle 0^\circ) (0.5084\angle 8.858^\circ) \\ &= 2.54\angle 8.858^\circ \text{ A}\end{aligned}$$

Represent in time domain.

$$i_o(t) = 2.54 \sin(t + 8.858^\circ) \text{ A}$$

Therefore, the value of the current $i_o(t)$ is $\boxed{2.54 \sin(t + 8.858^\circ) \text{ A}}$.

Step 7 of 15

(b)

For $\omega = 5 \text{ rad/s}$

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute 5 rad/s for ω and 2 H for L .

$$\begin{aligned}X_L &= j(5)(2) \\ &= j10\Omega\end{aligned}$$

Therefore, the value of the reactance X_L is $j10\Omega$.

Calculate the value of the reactance due to capacitance.

$$X_C = \frac{1}{j\omega C}$$

Step 8 of 15

Substitute 5 rad/s for ω and 50 mF for C .

$$\begin{aligned}X_C &= \frac{1}{j(5)(50 \times 10^{-3})} \\ &= -j4\Omega\end{aligned}$$

Therefore, the value of the reactance X_C is $-j4\Omega$.

Step 9 of 15

Draw the circuit diagram in phasor representation as shown in Figure2.

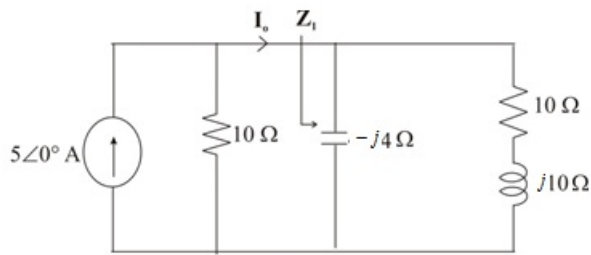


Figure 2

Step 10 of 15

Calculate the value of the impedance Z_1 .

$$\begin{aligned} Z_1 &= -j4 \parallel (10 + j10) \\ &= \frac{(-j4)(10 + j10)}{(-j4) + (10 + j10)} \\ &= \frac{40 - j40}{10 + j6} \\ &= 1.176 - j4.70588 \Omega \end{aligned}$$

Step 11 of 15

Calculate the value of the current I_o using current division rule.

$$I_o = (5\angle 0^\circ) \left(\frac{10}{Z_1 + 10} \right)$$

Substitute $1.176 - j4.70588 \Omega$ for Z_1 .

$$\begin{aligned} I_o &= (5\angle 0^\circ) \left(\frac{10}{1.176 - j4.70588 + 10} \right) \\ &= (5\angle 0^\circ) \left(\frac{10}{11.176 - j4.70588} \right) \\ &= (5\angle 0^\circ) (0.824 \angle 22.83^\circ) \\ &= 4.12 \angle 22.83^\circ \text{ A} \end{aligned}$$

Represent in time domain.

$$i_o(t) = 4.12 \sin(5t + 22.83^\circ) \text{ A}$$

Therefore, the value of the current $i_o(t)$ is $4.12 \sin(5t + 22.83^\circ) \text{ A}$.

Step 12 of 15

(c)

For $\omega = 10 \text{ rad/s}$

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute 10 rad/s for ω and 2 H for L .

$$\begin{aligned} X_L &= j(10)(2) \\ &= j20 \Omega \end{aligned}$$

Therefore, the value of the reactance X_L is, $j20 \Omega$.

Step 13 of 15

Calculate the value of the reactance due to capacitance.

$$X_C = \frac{1}{j\omega C}$$

Substitute 10 rad/s for ω and 50 mF for C .

$$\begin{aligned} X_C &= \frac{1}{j(10)(50 \times 10^{-3})} \\ &= -j2 \Omega \end{aligned}$$

Therefore, the value of the reactance X_C is, $-j2 \Omega$.

Step 14 of 15

Draw the circuit diagram in phasor representation as shown in Figure 3.

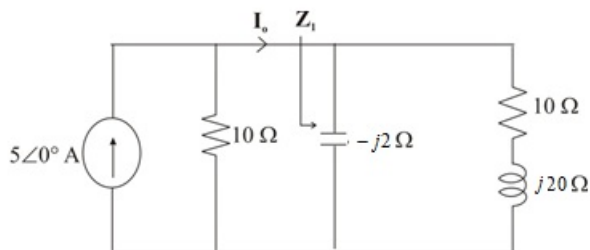


Figure 3

Calculate the value of the impedance Z_1 .

$$\begin{aligned} Z_1 &= -j2 \parallel (10 + j20) \\ &= \frac{(-j2)(10 + j20)}{(-j2) + (10 + j20)} \\ &= \frac{40 - j20}{10 + j18} \\ &= 0.0943 - j2.1698 \Omega \end{aligned}$$

Step 15 of 15

Calculate the value of the current $\mathbf{I}_o$ using current division rule.

$$\mathbf{I}_o = (5\angle 0^\circ) \left(\frac{10}{\mathbf{Z}_1 + 10} \right)$$

Substitute $0.0943 - j2.1698\Omega$ for $\mathbf{Z}_1$.

$$\begin{aligned}\mathbf{I}_o &= (5\angle 0^\circ) \left(\frac{10}{0.0943 - j2.1698 + 10} \right) \\ &= (5\angle 0^\circ) \left(\frac{10}{10.0943 - j2.1698} \right) \\ &= (5\angle 0^\circ) (0.968\angle 12.131^\circ) \\ &= 4.8242\angle 12.131^\circ \text{ A}\end{aligned}$$

Represent in time domain.

$$i_o(t) = 4.824\sin(10t + 12.131^\circ) \text{ A}$$

Therefore, the value of the current $i_o(t)$ is, $\boxed{4.824\sin(10t + 12.131^\circ) \text{ A}}$.